



Department of Computer Science and Engineering
Green University of Bangladesh
Midterm Examination, Fall 2024
CSE 303, Microprocessors and Microcontrollers



Duration: 1.5 Hours

Total Marks: 30

Answer all of the following questions.

(The CO# along with each sub-question represents the mapping of the questions onto the expected outcomes of the course provided in the course outline.)

Question 1.

- a. CO1 1+3 Marks
Explain the purpose of the stack in a microprocessor. Describe the steps involved in the PUSH PSW operation, along with an appropriate diagram.

— OR —

Critically compare the internal architecture features of the 8085 and 8086 microprocessors. Highlight the distinctive characteristics that set these two microprocessors apart in terms of their internal structures.

- b. CO1 3 Marks
✓ Predict the ending address of 8KB memory with the starting address 4A00 H.
- c. CO1 3 Marks
Draw the timing and state diagram for the opcode fetch cycle in the operation of the 8085 microprocessor.

Question 2.

- a. CO2 3 Marks
✓ Suppose a 2K*8 ROM is connected with an 8085 microprocessor from an address range of 5000H to 57FFH. Detect the chip select logic for the given ROM.

— OR —

- ✓ Detect the number of 256x8 ROM chips required to design 16KB of memory.
- b. CO2 2 Marks
Consider the below states of the microprocessor pins. Analyze what this configuration indicates:

$$\begin{aligned}\overline{IO/M} + \overline{RD} &= 1 + 0 \\ \overline{IO/M} + \overline{WR} &= 0 + 1\end{aligned}$$

C. **CO2** 5 Marks

Determine the count value, N, for the following code to produce a time delay of 0.5 milliseconds in a microprocessor with an 8 MHz quartz crystal clock source.

```
CALL addr16
MVI C,0
MVI D,N
LOOP: DCR D
ICR C
JNZ LOOP
RET
```

where, the T states required for each instruction are as follows-

Instructions	T-State required for execution of an instruction
CALL addr16	18
MVI C,0	7
MVI D,N	7
DCR D	4
ICR C	4
JNZ LOOP	10 (or 7)
RET	10

Question 3.

a. **CO1** 3 Marks

Show the status of the flag register after executing the following instruction line by line-

```
MVI A, 2E H
MVI B, F3 H
ADD B
HLT
```

b. **CO1** 3 Marks

Given the binary data 10011100 with the initial Carry Flag (CY) set to 1, perform the following operations:

- Apply the RAR instruction twice.
- Apply the RLC instruction once on the result obtained after the second RAR operation

After each operation, show the resulting data and the status of the Carry Flag (CY).

c. **CO1** 4 Marks

Discuss the different addressing modes of 8085 microprocessors with examples.

— OR —

Differentiate between maskable and non-maskable interrupts in 8085 microprocessors and explain how the 8085 microprocessor handles multiple interrupts occurring simultaneously.

□ End of Question □